Analysis Report

Assignment 5

We are interested in variety of selected features in the response of variable ***Bush in election2000.csv data***, the percentage of a state's total counted vote that was for George Bush in the 2000 presidential election, based on a subset of the following independent variables:

UnE = the unemployment rate for the state
Pop = the total state population
Male = the proportion of males in the state
Male18 = the proportion of males older than 18
Pop65 = the proportion of the population older than 65
NonMet = the proportion of rural (non metro) population
Pov = the proportion of the population below the poverty rate
NuHouse = the total number of households
Inc50 = the proportion of households earning more than $50,000
Inc75 = the proportion of households earning more than $75,000
Inc100 = the proportion of households earning more than $100,000

1. Fit a multivariate linear regression with all predictors (model 1). Report the fitted model, and the corresponding adjusted R-square, AIC, BIC. Present the predictors which does not have significant impact on the dependent variable Bush given 0.05 significance level.

Step 1: Fit a multivariate linear regression with all predictors

```
import statsmodels.api as sm
ols = sm.OLS(data.iloc[:,0], sm.add_constant(data.iloc[:,1:]))
lm1 = ols.fit()
print(lm1.summary())
```

Step 2: Reporting information

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   Bush   R-squared:                       0.815
Model:                            OLS   Adj. R-squared:                  0.755
Method:                 Least Squares   F-statistic:                     13.61
Date:                Tue, 24 Feb 2026   Prob (F-statistic):           1.89e-09
Time:                        19:23:42   Log-Likelihood:                -133.98
No. Observations:                  46   AIC:                             292.0
Df Residuals:                      34   BIC:                             313.9
Df Model:                          11
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       -652.3175    191.613     -3.404      0.002   -1041.722    -262.913
UnE           -2.0342      1.147     -1.774      0.085      -4.365       0.297
Pop        -3.453e-07   5.78e-07     -0.598      0.554   -1.52e-06    8.29e-07
Male          54.3664     14.425      3.769      0.001      25.051      83.682
Male18       -40.3098     11.056     -3.646      0.001     -62.779     -17.841
```

```
Pop65         -1.2891      0.543     -2.373      0.023      -2.393      -0.185
NonMet         0.2218      0.059      3.788      0.001       0.103       0.341
Pov            0.2802      0.508      0.552      0.585      -0.752       1.312
NuHouse     1.498e-06   1.65e-06      0.909      0.370   -1.85e-06    4.85e-06
Inc50         -0.0557      0.661     -0.084      0.933      -1.399       1.288
Inc75          0.0249      1.666      0.015      0.988      -3.361       3.411
Inc100        -1.2586      2.106     -0.598      0.554      -5.538       3.021
==============================================================================
Omnibus:                        0.436   Durbin-Watson:                   2.386
Prob(Omnibus):                  0.804   Jarque-Bera (JB):                0.549
Skew:                           0.204   Prob(JB):                        0.760
Kurtosis:                       2.654   Cond. No.                     2.12e+09
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 2.12e+09. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The fitted regression equation is:

- Bush = -652.3175 – 2.0342UnE – 3.453*10^{-7} Pop + 54.3664Male – 40.3098Male18 – 1.2891Pop65 + 0.2218NonMet + 0.2802Pov + 1.498*10^{-6}NuHouse – 0.0557Inc50 + 0.0249Inc75 – 1.2586Inc100

Model Statistics:

- R-squared adjusted: 0.755
- AIC: 292.0
- BIC: 313.9

The predictors which are not significant in predicting the response value Bush with p-value greater than 0.05 are: UnE, Pop, Pov, NuHouse, Inc50, Inc75, Inc100

2. Apply best subset approach for variable selection (model 2). We choose BIC as our criterion. Present which features are selected in the final model. Also report the fitted model, and the corresponding adjusted R-square, AIC, BIC.

Step 1: Define the function fit_lm() with the input as a set of selected features and output be the fitted model with corresponding RSS, R-squared adjusted, AIC, and BIC.

```
import itertools
def fit_lm(feature_set):
    ols = sm.OLS(data.Bush, sm.add_constant(data[list(feature_set)]))
    lm = ols.fit()
    RSS = ((lm.predict(sm.add_constant(data[list(feature_set)])) - data.Bush) ** 2).sum()
    rsquared_adj = lm.rsquared_adj
    AIC = lm.aic
    BIC = lm.bic
    return {"model":lm, "RSS":RSS,"Rsquare_adj": rsquared_adj,"AIC":AIC,"BIC":BIC}
```

Step 2: Define the function getbest() with the unput as the number of the selected features and the output as the best model given the pre-specified number of features. The best model refers to the one with lowest BIC as they are all equally complicated

```
def getbest(k):
    result = []
    for combo in itertools.combinations(data.columns[1:], k):
        result.append(fit_lm(combo))
    models = pd.DataFrame(result)
    best_model = models.loc[models["BIC"].argmin()]
    return best_model
```

Step 3: Find the best model with 1-12 features, and report corresponding RSS, R-squared adj, AIC, BIC

```
import time
models_best = pd.DataFrame(columns=["model","RSS","Rsquared_adj","AIC","BIC"])
start = time.time()
for i in range(1,len(data.columns)):
    models_best.loc[i] = getbest(i)
end = time.time()
print(models_best)
print("Best subset selection takes: ", end-start, "seconds")
best_index2 = models_best["BIC"].idxmin()
lm2 = models_best.loc[best_index2,"model"]
print(lm2.summary())
```

```
                                                 model          RSS  \
1   <statsmodels.regression.linear_model.Regressio...  2969.238063
2   <statsmodels.regression.linear_model.Regressio...  2166.632919
3   <statsmodels.regression.linear_model.Regressio...  1384.370173
4   <statsmodels.regression.linear_model.Regressio...  1166.068477
5   <statsmodels.regression.linear_model.Regressio...  1052.406013
6   <statsmodels.regression.linear_model.Regressio...    986.58975
7   <statsmodels.regression.linear_model.Regressio...   937.245641
8   <statsmodels.regression.linear_model.Regressio...    922.92846
9   <statsmodels.regression.linear_model.Regressio...   912.863033
10  <statsmodels.regression.linear_model.Regressio...   912.467606
11  <statsmodels.regression.linear_model.Regressio...   912.461597

   Rsquared_adj         AIC         BIC
1           NaN  326.243631  329.900914
2           NaN  313.747602  319.233526
3           NaN  295.142867  302.457432
4           NaN  289.248923   298.39213
5           NaN  286.531217  297.503065
6           NaN  285.560539  298.361028
7           NaN  285.200329  299.829461
8           NaN   286.49222  302.949993
9           NaN   287.98779  306.274204
10          NaN   289.96786  310.082915
11          NaN  291.967557  313.911254
Best subset selection takes:  9.368871688842773 seconds
                            OLS Regression Results
==============================================================================
Dep. Variable:                   Bush   R-squared:                       0.787
Model:                            OLS   Adj. R-squared:                  0.760
Method:                 Least Squares   F-statistic:                     29.48
Date:                Thu, 26 Feb 2026   Prob (F-statistic):           2.02e-12
Time:                        12:08:07   Log-Likelihood:                -137.27
No. Observations:                  46   AIC:                             286.5
Df Residuals:                      40   BIC:                             297.5
```

```
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       -717.4821    146.570     -4.895      0.000   -1013.712    -421.252
Male          60.5757     12.043      5.030      0.000      36.236      84.916
Male18       -45.3635      9.428     -4.811      0.000     -64.419     -26.308
Pop65         -1.0359      0.491     -2.110      0.041      -2.028      -0.043
NonMet         0.1794      0.044      4.089      0.000       0.091       0.268
Inc100        -1.7905      0.539     -3.324      0.002      -2.879      -0.702
==============================================================================
Omnibus:                        0.327   Durbin-Watson:                   2.305
Prob(Omnibus):                  0.849   Jarque-Bera (JB):                0.353
Skew:                           0.184   Prob(JB):                        0.838
Kurtosis:                       2.780   Cond. No.                     1.51e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 1.51e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The Male, Male18, Pop65, NonMet, and Inc100 features are selected in the final model, with the all p-values are 0.000, indicating they are significant in this model to predict Bush variable.

The fitted regression equation is:

- Bush = -717.4821 + 60.5757Male – 45.3635Male18 – 1.0359Pop65 + 0.1794NonMet – 1.7905Inc100

Model Statistics:

- R-squared adjusted: 0.760
- AIC: 286.5
- BIC: 297.5

3. Apply forward approach for variable selection (model 3). We choose BIC as our criterion. Present features are selected in the final model. Also report the fitted model, and the corresponding adjusted R-square, AIC, BIC.

Step 1: Create two lists to add each feature each time and assign starting time of function

```
remain = list(data.columns[1:])
picked = list()
model_best = pd.DataFrame(columns=["model","RSS","Rsquared_adj","AIC","BIC"])
start = time.time()
```

Step 2: Construct forward approach for variable selection

```
for i in range(1,len(data.columns)):
    best_BIC = np.inf
    for combo in itertools.combinations(remain,1):
        BIC = fit_lm(list(combo) + picked)["BIC"]
        if BIC < best_BIC:
            best_BIC = BIC
            best_feature = combo[0]
            model_best.loc[i] = fit_lm(list(combo) + picked)
    picked.append(best_feature)
    remain.remove(best_feature)
```

Step 3: Assign ending time and display information on the screen

```
end = time.time()
best_index3 = model_best1["BIC"].idxmin()
lm3 = model_best1.loc[best_index3,"model"]
print(model_best1)
print("Foward selection take: ", end-start, "seconds")
print(lm3.summary())
```

```
                                                 model          RSS  \
1   <statsmodels.regression.linear_model.Regressio...  2969.238063
2   <statsmodels.regression.linear_model.Regressio...  2356.629542
3   <statsmodels.regression.linear_model.Regressio...  1384.370173
4   <statsmodels.regression.linear_model.Regressio...  1166.068477
5   <statsmodels.regression.linear_model.Regressio...  1060.690673
6   <statsmodels.regression.linear_model.Regressio...   986.589750
7   <statsmodels.regression.linear_model.Regressio...   938.527163
8   <statsmodels.regression.linear_model.Regressio...   928.362027
9   <statsmodels.regression.linear_model.Regressio...   922.062471
10  <statsmodels.regression.linear_model.Regressio...   912.651987
11  <statsmodels.regression.linear_model.Regressio...   912.461597

    Rsquared_adj         AIC         BIC
1            NaN  326.243631  329.900914
2            NaN  317.614276  323.100200
3            NaN  295.142867  302.457432
4            NaN  289.248923  298.392130
5            NaN  286.891916  297.863765
6            NaN  285.560539  298.361028
7            NaN  285.263184  299.892315
8            NaN  286.762242  303.220015
9            NaN  288.449038  306.735452
10           NaN  289.977154  310.092209
11           NaN  291.967557  313.911254
Foward selection take:  0.5218911170959473 seconds

                            OLS Regression Results
==============================================================================
Dep. Variable:                   Bush   R-squared:                       0.785
Model:                            OLS   Adj. R-squared:                  0.758
Method:                 Least Squares   F-statistic:                     29.19
Date:                Thu, 26 Feb 2026   Prob (F-statistic):           2.35e-12
Time:                        12:08:08   Log-Likelihood:                -137.45
No. Observations:                  46   AIC:                             286.9
Df Residuals:                      40   BIC:                             297.9
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
```

```
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       -775.2704    141.775     -5.468      0.000   -1061.809    -488.732
Pop65         -0.9741      0.489     -1.993      0.053      -1.962       0.013
NonMet         0.1785      0.044      4.029      0.000       0.089       0.268
Male          64.6501     11.764      5.495      0.000      40.874      88.426
Male18       -48.3254      9.239     -5.231      0.000     -66.998     -29.652
Inc75         -0.7769      0.238     -3.263      0.002      -1.258      -0.296
==============================================================================
Omnibus:                        0.114   Durbin-Watson:                   2.276
Prob(Omnibus):                  0.945   Jarque-Bera (JB):                0.281
Skew:                           0.094   Prob(JB):                        0.869
Kurtosis:                       2.666   Cond. No.                     1.46e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 1.46e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The Pop65, NonMet, Male, Male18, Inc75 features are selected in the final model, with the p-value are less than 0.05 - except Pop65 is marginally insignificant at the 0.05 level, indicating they are significant in this model to predict Bush variable.

The fitted regression equation is:

- Bush = -775.2704 – 0.9741Pop65 + 0.1785NonMet + 64.6501Male – 48.3254Male18 – 0.7769Inc75

Model Statistics:

- R-squared adjusted: 0.758
- AIC: 286.9
- BIC: 297.9

4. Apply backward approach for variable selection (model 4). We choose BIC as our criterion. Present features are selected in the final model and report the fitted model, and the corresponding adjusted R-square, AIC, BIC.

Step 1: Create a list to remove a feature each time and assign starting time of function

```
pick = list(data.columns[1:])
model_best = pd.DataFrame(columns=["model","RSS","Rsquared_adj","AIC","BIC"])
start = time.time()
```

Step 2: Construct backward approach for variable selection

```
for i in range(1,len(data.columns)):
    best_BIC = np.inf
    for combo in itertools.combinations(pick,len(pick)-1):
        BIC = fit_lm(list(combo))["BIC"]
        if BIC < best_BIC:
            best_BIC = BIC
            best_feature = combo
            model_best.loc[i] = fit_lm(list(combo))
    pick = best_feature
```

Step 3: Assign ending time and display information on the screen

```
end = time.time()
best_index4 = model_best2["BIC"].idxmin()
lm4 = model_best2.loc[best_index4, "model"]
print(model_best2)
print("Backward selection takes: ", end-start, "seconds")
print(lm4.summary())
```

```
                                                 model          RSS  \
1   <statsmodels.regression.linear_model.Regressio...   912.467606
2   <statsmodels.regression.linear_model.Regressio...   912.863033
3   <statsmodels.regression.linear_model.Regressio...   922.928460
4   <statsmodels.regression.linear_model.Regressio...   937.245641
5   <statsmodels.regression.linear_model.Regressio...   987.109192
6   <statsmodels.regression.linear_model.Regressio...  1052.406013
7   <statsmodels.regression.linear_model.Regressio...  1169.495053
8   <statsmodels.regression.linear_model.Regressio...  1384.370173
9   <statsmodels.regression.linear_model.Regressio...  2348.014847
10  <statsmodels.regression.linear_model.Regressio...  3568.421052
11  <statsmodels.regression.linear_model.Regressio...  4930.727765

    Rsquared_adj         AIC         BIC
1            NaN  289.967860  310.082915
2            NaN  287.987790  306.274204
3            NaN  286.492220  302.949993
4            NaN  285.200329  299.829461
5            NaN  285.584751  298.385241
6            NaN  286.531217  297.503065
7            NaN  289.383899  298.527106
8            NaN  295.142867  302.457432
9            NaN  317.445814  322.931738
10           NaN  334.699252  338.356534
11           NaN  347.573967  349.402609
Backward selection takes:  0.4330761432647705 seconds

                            OLS Regression Results
==============================================================================
Dep. Variable:                   Bush   R-squared:                       0.787
Model:                            OLS   Adj. R-squared:                  0.760
Method:                 Least Squares   F-statistic:                     29.48
Date:                Thu, 26 Feb 2026   Prob (F-statistic):           2.02e-12
Time:                        12:08:08   Log-Likelihood:                -137.27
No. Observations:                  46   AIC:                             286.5
Df Residuals:                      40   BIC:                             297.5
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       -717.4821    146.570     -4.895      0.000   -1013.712    -421.252
Male          60.5757     12.043      5.030      0.000      36.236      84.916
Male18       -45.3635      9.428     -4.811      0.000     -64.419     -26.308
Pop65         -1.0359      0.491     -2.110      0.041      -2.028      -0.043
NonMet         0.1794      0.044      4.089      0.000       0.091       0.268
Inc100        -1.7905      0.539     -3.324      0.002      -2.879      -0.702
==============================================================================
```

```
Omnibus:                        0.327   Durbin-Watson:                   2.305
Prob(Omnibus):                  0.849   Jarque-Bera (JB):                0.353
Skew:                           0.184   Prob(JB):                        0.838
Kurtosis:                       2.780   Cond. No.                     1.51e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 1.51e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The Male, Male18, Pop65, NonMet, and Inc100 features are selected in the final model, with the all p-values are 0.000, indicating they are significant in this model to predict Bush variable.

The fitted regression equation is:

- Bush = -717.4821 + 60.5757Male – 45.3635Male18 – 1.0359Pop65 + 0.1794NonMet – 1.7905Inc100

Model Statistics:

- R-squared adjusted: 0.760
- AIC: 286.5
- BIC: 297.5

5. Apply forward approach for variable selection. This time, we determine the best model based on K-fold cross validation (you can decide the value of K) (model 5). Present features are selected in the final model and please report the fitted model, and the corresponding adjusted R-square, AIC, BIC.

Step 1: Get result from forward approach in question 3 stored in variable model_best1

Step 2: Using K-fold cross validation

```python
from sklearn.model_selection import KFold
from sklearn.metrics import mean_squared_error

cv = KFold(n_splits=5, random_state=1, shuffle=True)
mse = []

for i in range(1,12):
    x = data[model_best1.loc[i,"model"].params.index[1:]]
    y = data.iloc[:,0]
    error = []

    for train_index, test_index in cv.split(data):
        x_train, x_test = x.iloc[train_index], x.iloc[test_index]
        y_train, y_test = y.iloc[train_index], y.iloc[test_index]

        ols = sm.OLS(y_train, sm.add_constant(x_train))
        lm = ols.fit()
        y_pred = lm.predict(sm.add_constant(x_test))
        error.append(mean_squared_error(y_pred,y_test))
    mse.append(np.mean(error))

model_best1["MSE"] = mse
best_index4 = model_best1["MSE"].idxmin()
lm5 = model_best1.loc[best_index4, "model"]

print(lm5.summary())
```

Step 3: Present OLS table

```
OLS Regression Results
==============================================================================
Dep. Variable:                   Bush   R-squared:                       0.785
Model:                            OLS   Adj. R-squared:                  0.758
Method:                 Least Squares   F-statistic:                     29.19
Date:                Thu, 26 Feb 2026   Prob (F-statistic):           2.35e-12
Time:                        12:08:08   Log-Likelihood:                -137.45
No. Observations:                  46   AIC:                             286.9
Df Residuals:                      40   BIC:                             297.9
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       -775.2704    141.775     -5.468      0.000   -1061.809    -488.732
Pop65         -0.9741      0.489     -1.993      0.053      -1.962       0.013
NonMet         0.1785      0.044      4.029      0.000       0.089       0.268
Male          64.6501     11.764      5.495      0.000      40.874      88.426
Male18       -48.3254      9.239     -5.231      0.000     -66.998     -29.652
Inc75         -0.7769      0.238     -3.263      0.002      -1.258      -0.296
==============================================================================
Omnibus:                        0.114   Durbin-Watson:                   2.276
Prob(Omnibus):                  0.945   Jarque-Bera (JB):                0.281
Skew:                           0.094   Prob(JB):                        0.869
Kurtosis:                       2.666   Cond. No.                     1.46e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 1.46e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The Pop65, NonMet, Male, Male18, Inc75 features are selected in the final model, with the p-value are less than 0.05, except Pop65 (0.053), indicating they are significant in this model to predict Bush variable.

The fitted regression equation is:

- Bush = -775.2704 – 0.9741Pop65 + 0.1785NonMet + 64.6501Male – 48.3254Male18 – 0.7769Inc75

Model Statistics:

- R-squared adjusted: 0.758
- AIC: 286.9
- BIC: 297.9

6. Make prediction using all the models (model 1-5) that you just constructed on the test set election2000_test.csv, and report the mean square error (MSE).

Step 1: Import data from test file

Step 2: Define function to predict model and calculate mean square error

```
def predict_model(lm,test):
    predictors = lm.params.index[1:]

    x_test = test[predictors]
    x_test = sm.add_constant(x_test)

    y_test = test.iloc[:,0]

    y_pred = lm.predict(x_test)

    mse = mean_squared_error(y_test, y_pred)

    print(mse)
```

Step 3: Apply function to models 1-5 and present the result

```
print("MSE for model 1: ",predict_model(lm1,test))
print("MSE for model 2: ",predict_model(lm2,test))
print("MSE for model 3: ",predict_model(lm3,test))
print("MSE for model 4: ",predict_model(lm4,test))
print("MSE for model 5: ",predict_model(lm5,test))
```

```
MSE for model 1:  83.04050107781043
MSE for model 2:  68.97138242923516
MSE for model 3:  76.37542126209755
MSE for model 4:  68.97138242923516
MSE for model 5:  76.37542126209755
```

7. Compare all the regression models that you have constructed by creating a comparison table with the metrics: adjust R-square, AIC, BIC, and predicted MSE.

Model	Adjusted R-square	AIC	BIC	Predicted MSE
1	0.755	292.0	313.9	83.0405
2	0.760	286.5	297.5	68.9714
3	0.758	286.9	297.9	76.3754
4	0.760	286.5	297.5	68.9714
5	0.758	286.9	297.9	76.3754

Noticed from previous steps, we have two pairs of models that have the same fitted regression lines with the same features, which are model 2&4 and model 3&5. Moreover, the statistics AICs, BICs, and adjusted R-squares are slightly different for each model, indicating that model 2-5 have the almost the same power to explain the response variable Bush. Model 2 and 4 selected the same predictors and achieved the same statistics result, indicating strong agreement between best subset and backward selection. The same happens with model 3 and 5 with the agreement between forward and K-fold selection, but with a slightly different income variable (Inc75 instead of Inc100 like model 2 and 4), making their performance slightly worse than the other two models. Thus, the features combination in model 2&4 is more preferable since it returns lowest Predicted MSE on the test data, giving us to the conclusion of the model with the fitted regression line

Bush = -717.4821 + 60.5757Male – 45.3635Male18 – 1.0359Pop65 + 0.1794NonMet – 1.7905Inc100

will be the best model with significant features that are selected through different methods.